Neural Network-based Power Flow Model

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***Abstract*—Power flow analysis is used to evaluate the flow of electricity in the power system network. Power flow calculation is used to determine the steady-state variables of the system, such as the voltage magnitude / phase angle of each bus and the active/reactive power flow on each branch. The DC power flow model is a popular linear power flow model that is widely used in the power industry. Although it is fast and robust, it may lead to inaccurate line flow results for some transmission lines. Since renewable energy sources such as solar farm or offshore wind farm are usually located far away from the main grid, accurate line flow results on these critical lines are essential for power flow analysis due to the unpredictable nature of renewable energy. Data-driven methods can be used to partially addressed these inaccuracies by taking advantage of historical grid profiles. In this paper, a neural network (NN) model is trained to predict power flow results using historical power system data. Although the training process may take time, once trained, it is very fast to estimate line flows. A comprehensive performance analysis between the proposed NN-based power flow model and the traditional DC power flow model is conducted. It can be concluded that the proposed NN-based power flow model can find solutions quickly and more accurately than DC power flow model.**

***Index Terms*— DC Power Flow, Machine learning, Neural network, Power flow, Renewable energy, Transmission network.**

I. Introduction

For any power system, power flow is necessary to analyze the steady-state of the system. AC Power flow problems are usually solved using iterative methods such as Gauss-Seidel (GS) method and Newton-Raphson (NR). These methods can provide steady-state solutions for the system within specified accuracy boundaries. However, for a large-scale system, with hundreds of thousands of buses, it is not practical to use AC power flow equation-based methods for decision making or fast screening of the system, especially system with multiple solar or wind farms. Renewable sources provide clean, environmentally friendly energy to the power system. However, due to the inherent intermittent nature of renewable energy, novel solutions to mitigate uncertainties from renewable sources are desirable. Outside of the challenges of securing green and clean energy sources, the power industry must face difficult problems of integrating renewable energy sources to the existing aging power infrastructure. Problems will arise in terms of economic feasibility, reliability, and security. Applying ML to solve these challenging situations is critical for development of clean and green energy of the future. Applications of machine learning (ML) toward renewable energy have been widely researched and studied in recent years [1]- [2]. For example, using recursive neural networks, researchers had tried to forecast energy prices of renewable energy sources for day-ahead energy market [1]. Alternatively, using ML as an advanced algorithm to predict generation of renewable sources has been proposed [3] [4]. Deep reinforcement learning was investigated as a possible control strategy for power systems with multiple renewable energy sources [5].

Thus, it can be concluded that using machine learning as an alternative approach for solving problems involving renewable energy in power systems can be quite beneficial. Studies focusing on applications of ML show that ML algorithms could be used to gain an advantage over traditional methods in addressing various issues in power systems. Compared to traditional computational approaches, machine learning algorithms have an intrinsic generalization capability with greater computational efficiency and scalability [6]. Machine learning algorithms also have the ability to learn complex nonlinear input-output relationships, use sequential training procedures, and reflexively adapt themselves to the data.

Machine learning has been used as an algorithm selector that chooses between different power flow management algorithms depending on the state of the network [7]. Instead of directly using ML methods to solve for power flow, [7] looked at the performance benefits of using machine learning in choosing the best algorithms for power flow management. In [8], a neural network (NN) model was proposed to predict voltage magnitude and active power flow of the IEEE 14-bus system. Although it proves that NN model is an efficient method for calculating power flow, the results were only shown for a selected number of buses and branches. For [9], predictions using an NN model were made for voltage magnitude and phase angle at each bus. However, a thorough analysis regarding the performance of the NN model vs. the non-iterative DC power flow method was not conducted. Moreover, predictions for active power flow for each branch were not considered. The NN models studied in [8] and [9] were both trained and tested against only one system. This restriction does not show how well the NN model would adapt against different power systems with different bus-branch configurations. Due to the nature of machine learning algorithms to be able to reflexively adapt to different data/environments, these models did not fully explore the potential of machine learning in solving power flow.

In this paper, we developed an ML algorithm that can solve

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基于神经网络的潮流模型

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摘要——潮流分析用于评估电力系统网络中的电力流动。潮流计算用于确定系统的稳态变量，例如各母线的电压幅值/相角以及各支路的有功/无功功率。直流潮流模型是一种常用的线性潮流模型，在电力行业中应用广泛。尽管它速度快、鲁棒性强，但对于某些输电线路，其潮流计算结果可能不准确。由于太阳能发电场或海上风电场等可再生能源通常远离主电网，考虑到可再生能源的不可预测性，这些关键线路的精确潮流计算结果对于潮流分析至关重要。数据驱动方法可以利用历史电网数据来部分解决这些不准确性问题。本文利用历史电力系统数据训练了一个神经网络（NN）模型来预测潮流结果。虽然训练过程可能需要一些时间，但一旦训练完成，线路潮流的估计速度非常快。本文对所提出的基于神经网络的潮流模型和传统的直流潮流模型进行了全面的性能分析。由此可以得出结论，所提出的基于神经网络的潮流模型能够比直流潮流模型更快、更准确地找到解决方案。

索引词——直流潮流、机器学习、神经网络、潮流、可再生能源、输电网络。

一、引言

对于任何电力系统而言，潮流计算对于分析系统的稳态至关重要。交流潮流问题通常采用迭代方法求解，例如高斯-赛德尔迭代法（GS）和牛顿-拉夫逊迭代法（NR）。这些方法能够在指定的精度范围内提供系统的稳态解。然而，对于拥有数十万个母线的大型系统，使用基于交流潮流方程的方法进行决策或快速筛选并不实际，尤其是在包含多个太阳能或风力发电场的系统中。可再生能源为电力系统提供清洁、环保的能源。然而，由于可再生能源固有的间歇性，亟需新的解决方案来降低可再生能源带来的不确定性。除了保障绿色清洁能源供应的挑战之外，电力行业还必须面对将可再生能源并入现有老化电力基础设施的难题。这些问题将涉及经济可行性、可靠性和安全性等方面。应用机器学习来解决这些挑战性问题对于未来清洁绿色能源的发展至关重要。

近年来，机器学习（ML）在可再生能源领域的应用得到了广泛的研究[1]-[2]。例如，研究人员曾尝试利用递归神经网络预测日前能源市场中可再生能源的能源价格[1]。此外，也有研究提出将机器学习作为一种先进的算法来预测可再生能源的发电量[3][4]。深度强化学习也被研究作为一种适用于包含多种可再生能源的电力系统的潜在控制策略[5]。

因此，可以得出结论，将机器学习作为解决电力系统中可再生能源相关问题的替代方法，具有相当大的益处。专注于机器学习应用的研究表明，机器学习算法在解决电力系统中的各种问题方面，比传统方法更具优势。与传统计算方法相比，机器学习算法具有更强的泛化能力，计算效率和可扩展性更高[6]。机器学习算法还能够学习复杂的非线性输入输出关系，采用顺序训练程序，并能根据数据进行自我调整。

机器学习已被用作算法选择器，根据网络状态在不同的潮流管理算法之间进行选择[7]。[7]并未直接使用机器学习方法求解潮流，而是着眼于利用机器学习选择最佳潮流管理算法的性能优势。[8]提出了一种神经网络（NN）模型来预测 IEEE 14 节点系统的电压幅值和有功功率潮流。尽管该模型证明了 NN 模型是计算潮流的有效方法，但其结果仅针对选定的节点和支路进行了展示。[9]使用 NN 模型预测了每个节点的电压幅值和相角。然而，该研究并未对 NN 模型与非迭代直流潮流计算方法的性能进行全面分析，也未考虑每个支路的有功功率潮流预测。[8]和[9]中研究的 NN 模型均仅针对一个系统进行训练和测试。这种局限性无法展现 NN 模型在不同节点-支路配置的电力系统中的适应能力。由于机器学习算法的特性使其能够自发地适应不同的数据/环境，因此这些模型并没有充分挖掘机器学习在解决潮流问题方面的潜力。

本文中，我们开发了一种能够解决以下问题的机器学习算法。

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power flow solutions quickly and accurately using an NN model. Detailed research was performed regarding model selection and how to maximize the performance of the proposed NN model. The model was trained and tested against multiple systems to assess how accurate it could predict the outputs. Comparing to DC power flow model, its performance and effectiveness are evaluated and demonstrated.

The rest of the paper is organized as follows. Background analysis of power flow and machine learning are covered in section II. Section III introduces how the NN model was developed, and sample data were generated. Section IV provides analysis of the results and evaluates the performance of the proposed NN method. Section V concludes the paper and section VI describes possible future work.

II. Preliminaries

A. Power Flow

Power flow problem is represented by a set of nodal power balance equations that simply state that the sum of active and reactive power at a node must be equal to zeros respectively, which follows the law of conservation of energy [10]. The nodal power balance equations are listed as follows [11]

$$P_i - \sum_{k=1}^{N} |V_i|\,|V_k|(G_{ik}\cos\theta_{ik} + B_{ik}\sin\theta_{ik}) = 0 \quad (1)$$

$$Q_i - \sum_{k=1}^{N} |V_i|\,|V_k|(G_{ik}\sin\theta_{ik} - B_{ik}\cos\theta_{ik}) = 0 \quad (2)$$

where, P_i and Q_i are the active and reactive power injections at each node respectively. The summation terms represent the active or reactive power injections or withdrawals from the electrical network at a given node i. $|V_i|$ and $|V_k|$ are the voltage magnitude for the two end buses of a branch. G_{ik} and B_{ik} are the corresponding conductance and susceptance of a branch. The phase angle θ_{ik} is the difference in voltage phase angles of the two end buses of a branch.

Using traditional AC power flow (ACPF) methods such as NR method and GS method, an accurate steady-state solution can be found. But, due to the inherent and complex iterative nature of these algorithms, it is not beneficial for some online monitoring applications or to be integrated into optimization-based scheduling and dispatching models [12]. An alternative non-iterative method called DC power flow (DCPF) can be used in such cases. For this method, at all nodes, the voltage magnitude is assumed to be 1, and reactive power is ignored [13]. With DCPF, the solutions to steady-state active power flow can be found very quickly. However, the values may be very inaccurate due to the assumptions made earlier. Thus, it is desirable to develop a new method that can provide fast and accurate power flow solutions.

B. Machine Learning

Machine learning is a computer algorithm that can automatically improve/learn through experience by using historical data. Models will be built using sample training data to make predictions or decisions without being explicitly programmed.

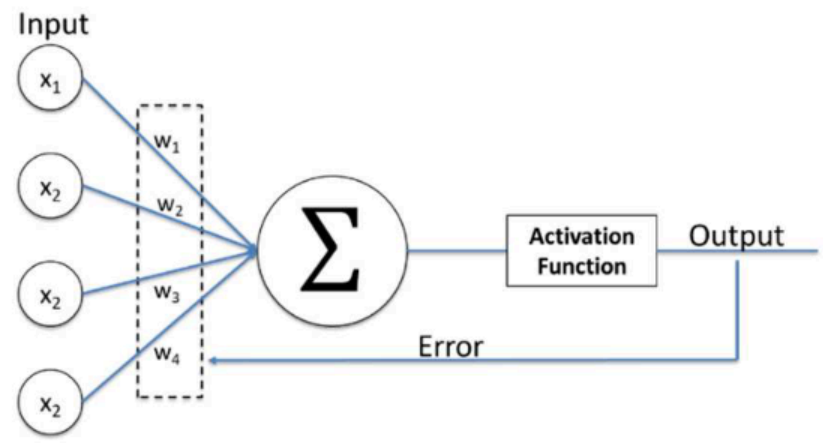


Fig. 1. Example of a simple machine learning model.

For a basic ML model in Fig. 1, training data (input) are multiplied first with a weight ($\boldsymbol{w}$) vector. Then, the results are mapped to an output value after applying an activation function. Depending on model selection, different activation functions can be used. During the training process, the $\boldsymbol{w}$ vector is repeatedly updated until the error is less than a targeted threshold or a specified number of epochs is reached [14].

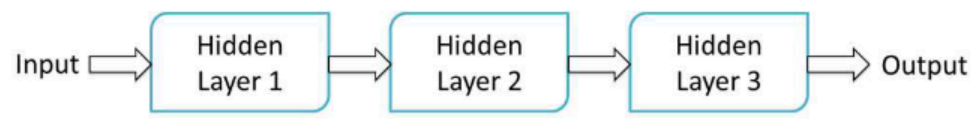


Fig. 2. Example of a multilayer neural network model.

An NN model typically involves multiple nodes (artificial neurons) with several hidden layers, such as the one from Fig. 2. The connections between nodes (vector weights) reflect the signal strength between each neuron. A complex NN with different layers performs different transformations of the input data. Each layer/node learns different features from the input data. An NN model can learn complex input-output relationships that can be difficult to comprehend or program using traditional algorithms.

The applications of ML methods are numerous across different industries, businesses, and research areas. One exciting application is to use artificial intelligence to automate the translation of text/speech. In addition, super-resolution enhancement of pictures is an interesting subject for machine learning [15]. As applications for machine learning are increasingly being explored and utilized, continuous development of machine learning is critical to solving existing and emerging challenging problems in power systems.

III. Model Selection

For the power flow problem studied in this paper, supervised learning through regression analysis was selected as the ML method to train the NN model with a large amount of data. The matpower module for MATLAB was used to solve AC power flow problems under various power system conditions and data samples (power flow results) were collected. Initial voltage and demand were varied within ±10% of the base values. For each test power system, at least 10,000 samples were generated. The data set was divided into three groups: 80% for training, 10% for validation, and 10% for testing.

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本文利用神经网络模型快速准确地求解潮流问题。针对模型选择以及如何最大化所提出的神经网络模型的性能，进行了深入研究。该模型经过训练，并在多个系统上进行了测试，以评估其预测输出的准确性。通过与直流潮流模型的比较，评估并验证了该模型的性能和有效性。

本文余下部分组织如下：第二节概述了潮流计算和机器学习的背景知识；第三节介绍了神经网络模型的开发过程以及样本数据的生成；第四节分析了结果并评估了所提出的神经网络方法的性能；第五节总结全文；第六节展望了未来的研究方向。

二、预备知识

A. 功率流

潮流问题可以用一组节点功率平衡方程来表示，这些方程简单地表明，节点处的有功功率和无功功率之和必须分别为零，这遵循能量守恒定律[10]。节点功率平衡方程如下所示[11]。

$$\sum \quad (1)$$

$$0 \quad (2)$$

其中，P 和 Q 分别表示各节点的有功功率和无功功率注入量。求和项表示给定节点从电网注入或提取的有功功率或无功功率。|V| 和 |V| 分别表示支路两端母线的电压幅值。G 和 Bar 分别表示支路的电导和电纳。相角表示支路两端母线电压相角的差值。

使用传统的交流潮流计算方法（例如 NR 法和 GS 法）可以得到精确的稳态潮流解。但是，由于这些算法固有的复杂迭代特性，它们并不适用于某些在线监测应用，也不便于集成到基于优化的调度模型中[12]。在这种情况下，可以使用一种称为直流潮流计算（DCPF）的非迭代方法。对于该方法，所有节点的电压幅值均假设为 1，并且忽略无功功率[13]。使用 DCPF 可以非常快速地找到稳态有功潮流解。然而，由于之前所做的假设，其结果可能非常不准确。因此，需要开发一种能够提供快速且精确潮流解的新方法。

B. 机器学习

机器学习是一种计算机算法，它能够利用历史数据，通过经验自动改进/学习。模型将使用样本训练数据构建，无需显式编程即可进行预测或决策。

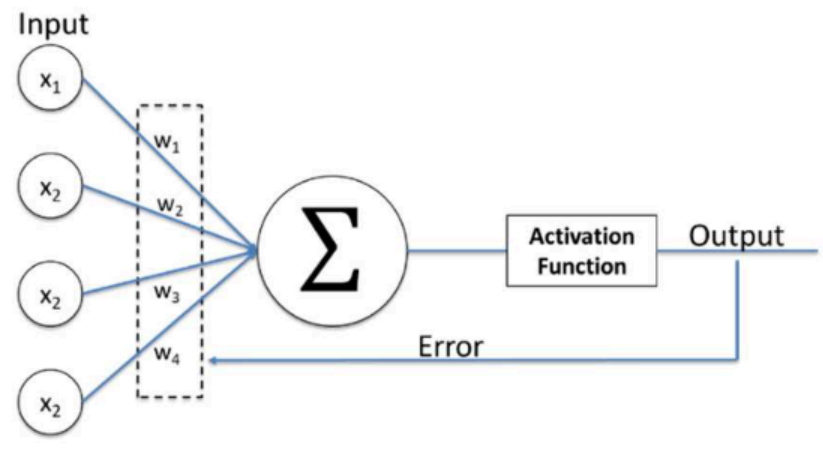


图 1. 一个简单的机器学习模型示例。

对于图 1 所示的基本机器学习模型，首先将训练数据（输入）与权重 (w) 向量相乘。然后，应用激活函数后，将结果映射到输出值。根据模型选择，可以使用不同的激活函数。在训练过程中，w 向量会不断更新，直到误差小于目标阈值或达到指定的训练轮数 [14]。

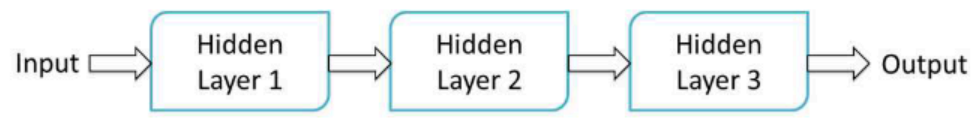


图 2. 多层神经网络模型示例。

神经网络模型通常包含多个节点（人工神经元）和若干隐藏层，如图 2 所示。节点之间的连接（向量权重）反映了每个神经元之间的信号强度。具有不同层的复杂神经网络会对输入数据进行不同的转换。每一层/节点都会从输入数据中学习不同的特征。神经网络模型可以学习复杂的输入输出关系，而这些关系使用传统算法可能难以理解或编程实现。

机器学习方法在各个行业、企业和研究领域都有着广泛的应用。其中一个令人兴奋的应用是利用人工智能实现文本/语音的自动翻译。此外，图像超分辨率增强也是机器学习的一个重要研究方向[15]。随着机器学习应用的不断探索和利用，机器学习的持续发展对于解决电力系统中现有和新兴的挑战性问题至关重要。

三、型号选择

本文研究的潮流问题，选择基于回归分析的监督学习作为机器学习方法，利用大量数据训练神经网络模型。使用 MATLAB 的 matpower 模块求解不同电力系统工况下的交流潮流问题，并采集数据样本（潮流计算结果）。初始电压和需量在基准值的±10%范围内变化。每个测试电力系统至少生成 10000 个样本。数据集被分为三组：80%用于训练，10%用于验证，10%用于测试。

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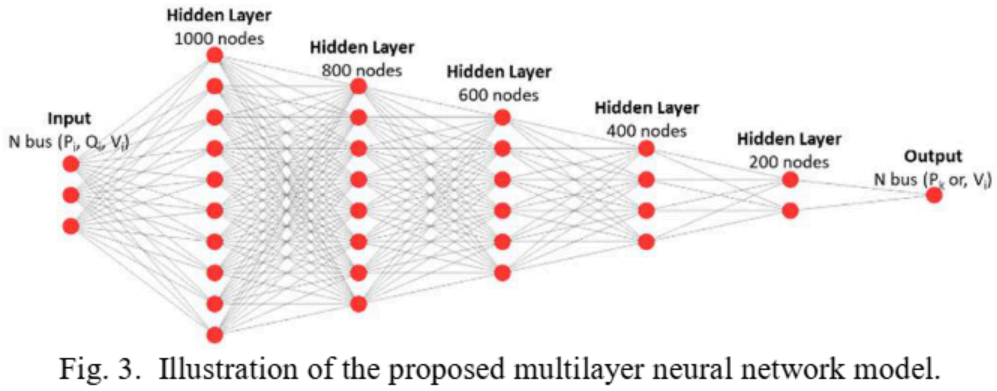


Fig. 3. Illustration of the proposed multilayer neural network model.

The proposed NN model as shown in Fig. 3 has five hidden layers. The Leaky Rectified Linear Unit (Leaky ReLU), as shown in Fig. 4, was chosen as the activation function for backpropagation. It allows a small, non-zero constant gradient to pass through to minimize the vanishing gradient problem during the training phase of a NN model.

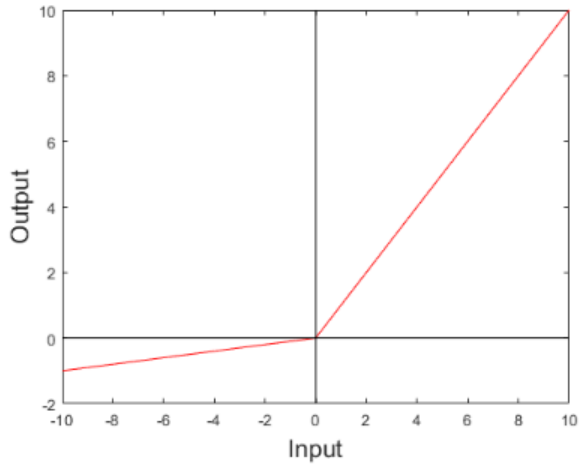


Fig. 4. Plot of leaky RELU function.

Input data for the proposed NN model include the initial voltage magnitude, the sum of active power injection, and reactive power injection for each bus. The output data for training will be steady-state voltage magnitude for each bus and active power flow for each branch.

Before feeding input data into the model for training, the data are normalized using the min-max normalization method. For every data feature, the minimum value is 0 and the maximum value is 1.

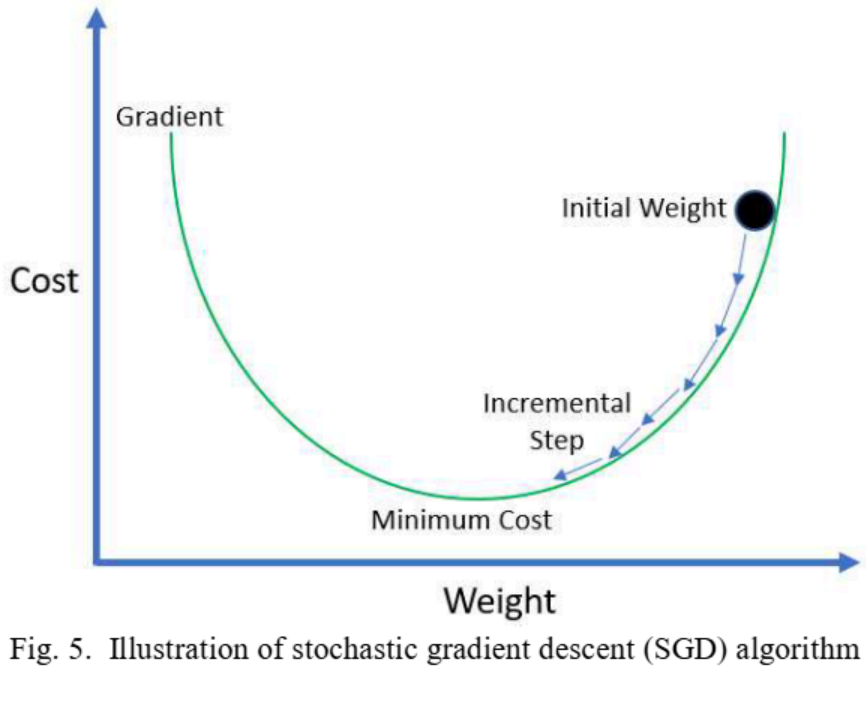


Fig. 5. Illustration of stochastic gradient descent (SGD) algorithm

Stochastic gradient descent (SGD) was chosen as the optimizing function for the NN model. SGD is very common for ML algorithm. In Fig. 5, a random starting point was chosen on the gradient curve of the objective cost function. At each subsequent iteration, the gradient was updated until the global minimum/threshold has been reached. Regular stochastic gradient requires computation of all data points for each iteration. For SGD, the selection of random samples in each minibatch for training can minimize computing time compared to using a complete training data set. Thus, it is desirable to use SGD as an optimization function.

Mean square error (MSE) loss was used to measure the performance of the model during the training process. MSE loss is commonly used for regression training of machine learning. The MSE loss function used in this paper is defined in (3), where y_i denotes the actual output value while $\tilde{y}_i$ denotes the NN model estimated output value, and n denotes the number of sample points.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \tilde{y}_i)^2 \tag{3}$$

IV. Results Analysis

The proposed NN model was trained to predict voltage magnitude and active power flow on multiple systems with different sizes: the IEEE 9-bus test system, IEEE 24-bus test system, IEEE 39-bus test system, IEEE 57-bus test system, and the IEEE 118-bus test system.

For comparison between the proposed NN model and DCPF model, absolute value difference and percent relative difference are used as evaluation metrics. Absolute value difference is defined as the Mean Absolute Error (MAE) between the target value against a reference value. The target values for comparison are the NN model's predictions and DCPF results. The reference base value is obtained from the full ACPF model.

$$MAE = \frac{\sum_{i=1}^{n}|x_i - y_i|}{n} \tag{4}$$

where x is the target value and y is the reference value.

Percent relative difference (PRD) was calculated using the relative difference formula shown below:

$$PRD = \frac{\sum_{i=1}^{n}\frac{2|x_i - y_i|}{|x_i + y_i|}}{n} * 100\% \tag{5}$$

This PRD metric is a dimensionless unit. The penalty of having a much smaller reference value relative to the target value is minimized when dividing by the sum of both values.

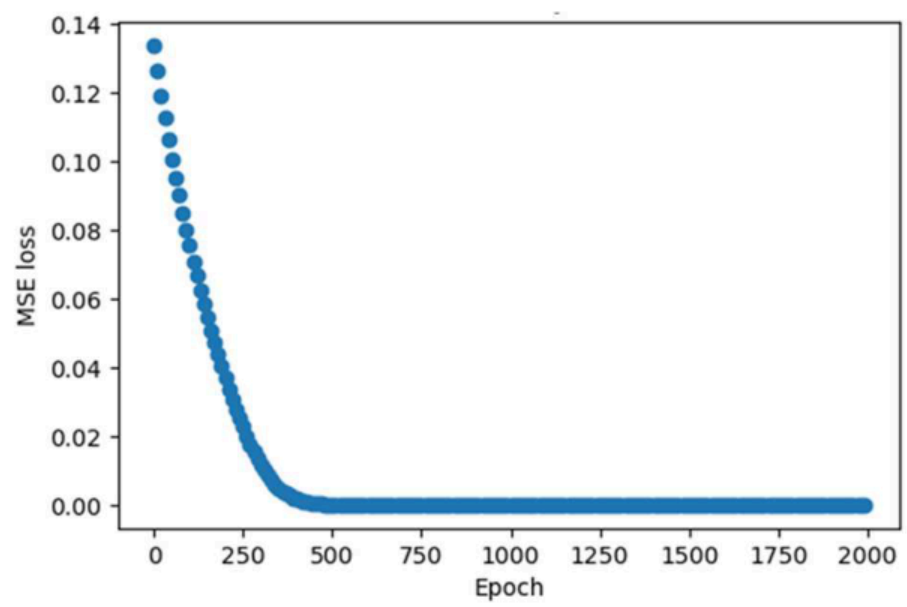


Fig. 6. MSE loss during training for active power flow of a 24-bus system.

In Fig. 6, the proposed NN model performs well in minimizing the MSE loss. The training loss decreases rapidly and then reaches the plateau zone after around 400 epochs. In addition, the model did an excellent job in fitting the predicted

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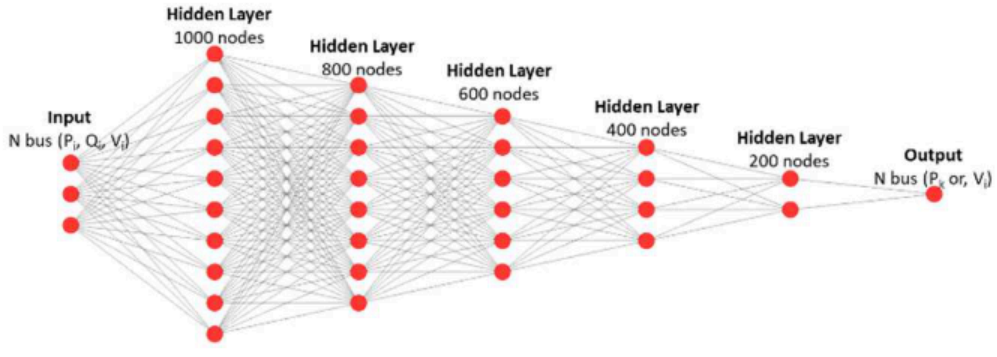


图 3. 所提出的多层神经网络模型示意图。

如图 3 所示，所提出的神经网络模型包含五个隐藏层。如图 4 所示，反向传播的激活函数选用漏失修正线性单元（Leaky ReLU）。它允许较小的非零梯度通过，从而最大限度地减少神经网络模型训练阶段的梯度消失问题。

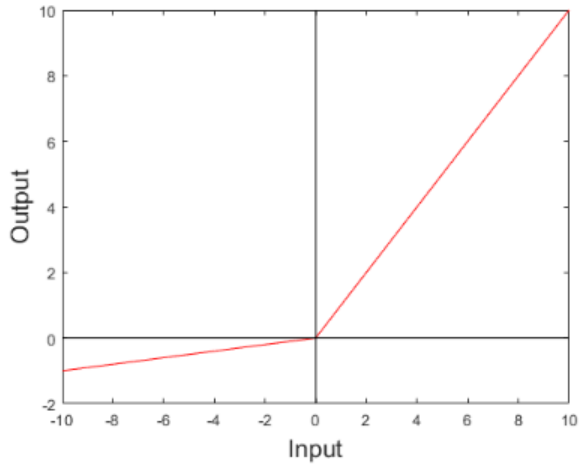


图 4. 漏电 RELU 函数图。

所提出的神经网络模型的输入数据包括各母线的初始电压幅值、有功功率注入总和以及无功功率注入总和。训练输出数据为各母线的稳态电压幅值和各支路的有功功率流。

在将输入数据输入模型进行训练之前，使用最小-最大归一化方法对数据进行归一化。对于每个数据特征，最小值为 0，最大值为 1。

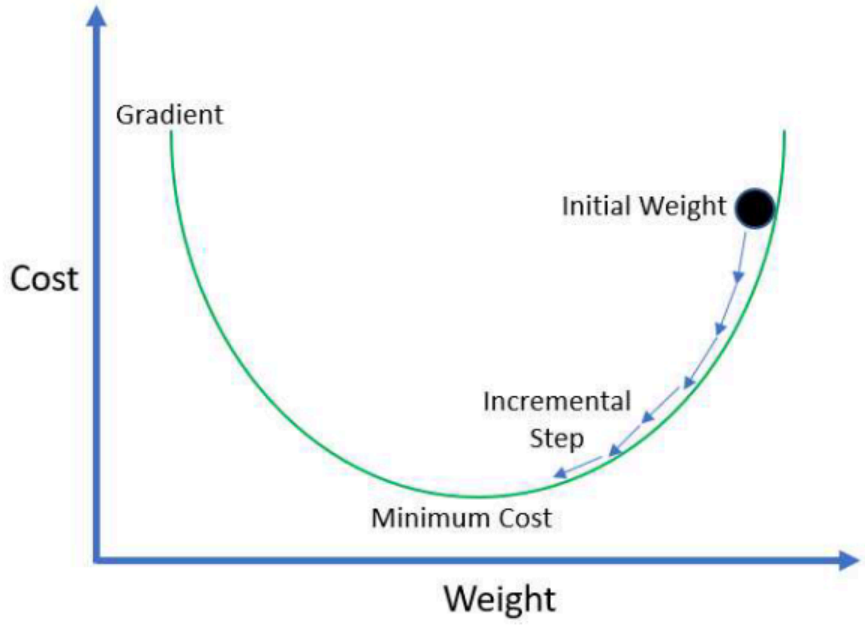


图 5. 随机梯度下降 (SGD) 算法示意图

随机梯度下降法（SGD）被选为神经网络模型的优化函数。SGD 在机器学习算法中非常常见。如图 5 所示，在目标成本函数的梯度曲线上随机选择一个起始点。在每次后续迭代中……

梯度会不断更新，直到达到全局最小值/阈值。常规随机梯度下降法（SGD）每次迭代都需要计算所有数据点。而对于 SGD，在每个小批量训练中选择随机样本可以最大限度地减少计算时间，相比之下，使用完整的训练数据集更为理想。因此，SGD 更适合作为优化函数。

均方误差（MSE）损失用于衡量模型在训练过程中的性能。MSE 损失常用于机器学习的回归训练。本文使用的 MSE 损失函数定义如式(3)所示，其中表示实际输出值，表示神经网络模型估计的输出值，n 表示样本点数。

Σ – (3)

四、结果分析

所提出的 NN 模型经过训练，可以预测不同规模的多个系统的电压幅值和有功功率潮流：IEEE 9 节点测试系统、IEEE 24 节点测试系统、IEEE 39 节点测试系统、IEEE 57 节点测试系统和 IEEE 118 节点测试系统。

为了比较所提出的神经网络模型和 DCPF 模型，本文采用绝对值差异和相对百分比差异作为评估指标。绝对值差异定义为目标值与参考值之间的平均绝对误差（MAE）。用于比较的目标值是神经网络模型的预测值和 DCPF 模型的结果。参考基准值则来自完整的 ACPF 模型。

重试 错误原因 (4)

其中 x 为目标值，y 为参考值。

百分比相对差异（PRD）采用以下相对差异公式计算：

（5）该 PRD 指标是一个无量纲单位。当参考值远小于目标值时，除以两个值之和可以最大限度地减少其带来的负面影响。

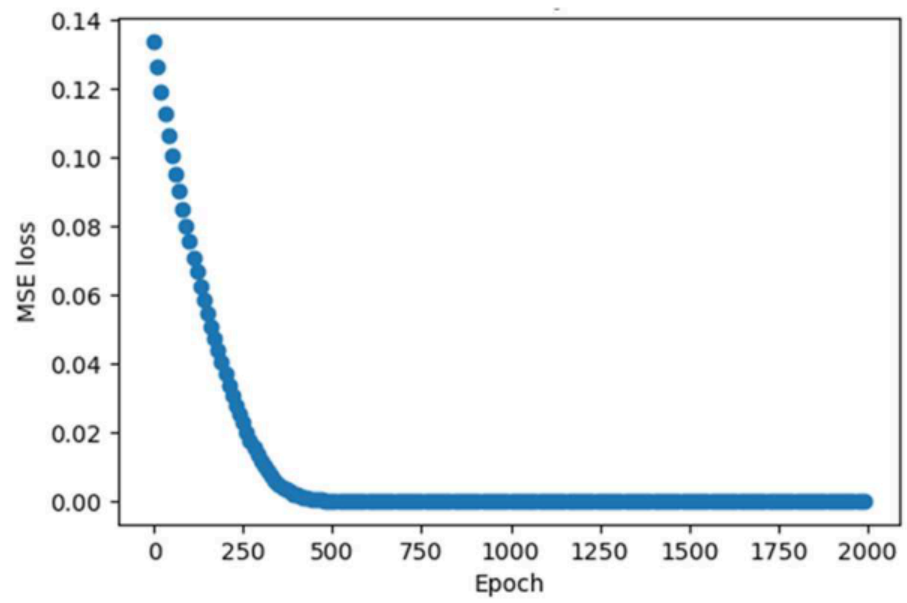


图 6. 24 节点系统有功功率潮流训练期间的 MSE 损失。

如图 6 所示，所提出的神经网络模型在最小化均方误差 (MSE) 损失方面表现良好。训练损失迅速下降，并在大约 400 个 epoch 后达到平台期。此外，该模型在拟合预测结果方面也表现出色。

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value toward the output value. The validation error rate curve nearly matched the training error rate curve in Fig. 7. The trained NN model does not seem to suffer from being underfitted or overfitted.

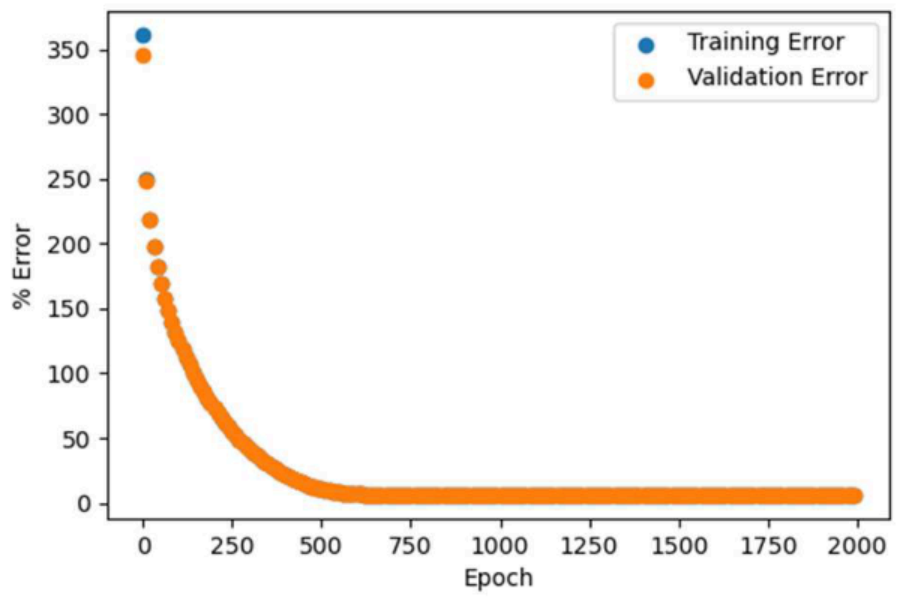


Fig. 7. MAE during training for active power flow of a 24-bus system

The proposed NN model predicts voltage magnitude at each bus for multiple systems with high accuracy: the error is around 0.5% as shown in Table I. However, since nearly all voltage magnitude values fall between a narrow range from 0.9 p.u. to 1.1 p.u., it was considerably easier for the NN model to learn and predict the output values.

TABLE I
Mean PRD of voltage magnitude of the proposed NN model on different test systems.

Voltage Magnitude	
System	**Percent Error**
9 Bus	0.54
24 Bus	0.60
39 Bus	0.50
57 Bus	0.33
118 Bus	0.59

Table II shows the absolute errors (in MW) for line flows estimated with the proposed NN model and the traditional DCPF model. The active line power flow predicted by the proposed NN model is much more accurate than the traditional DCPF model. The lines flows (MW) predicted with the proposed NN model are much closer to the reference values from the full exact ACPF model across all measured statistical values.

TABLE II
Absolute value (MW) difference of active power flow comparing the proposed NN model and the traditional DCPF model on different test systems.

System	Model	Mean	Max	Min	Median	Std.Dev.
9 Bus	**NN**	**1.70**	**6.54**	**0.00**	**1.50**	**1.22**
	DCPF	2.96	14.52	0.00	2.53	2.21
24 Bus	**NN**	**2.77**	**21.60**	**0.00**	**2.26**	**2.27**
	DCPF	5.92	43.38	0.00	4.29	5.53
39 Bus	**NN**	**7.07**	**45.69**	**0.00**	**5.27**	**6.65**
	DCPF	13.33	100.86	0.00	9.93	12.20
57 Bus	**NN**	**0.54**	**11.08**	**0.00**	**0.27**	**0.73**
	DCPF	1.65	23.19	0.00	0.96	1.98
118 Bus	**NN**	**1.16**	**20.86**	**0.00**	**0.84**	**1.27**
	DCPF	3.43	70.62	0.00	2.02	5.27

Table III presents the results associated with only highly loaded branches using different threshold levels. The DCPF results are not as accurate as the proposed NN model. For example, for all branches with actual active power flow over 200 MW for the IEEE 24-bus system, the PRD mean for the proposed NN model is only 0.65%, which is very close to the actual results reported from the full ACPF model. However, the PRD is 1.35% for the DCPF model. It is double the error of the proposed NN model.

TABLE III
PRD of active power flow comparing the proposed NN model and the DCPF model for different threshold levels on the IEEE 24-bus system.

System	Model	Mean	Max	Min	Median	Std.Dev.
50 MW	**NN**	**1.72**	**17.14**	**0.00**	**1.26**	**1.62**
	DCPF	6.42	75.28	0.00	3.43	10.02
100 MW	**NN**	**1.42**	**8.95**	**0.00**	**0.89**	**1.52**
	DCPF	6.55	75.28	0.00	2.75	12.76
150 MW	**NN**	**1.33**	**8.95**	**0.00**	**0.76**	**1.59**
	DCPF	2.96	24.63	0.00	1.63	3.53
200 MW	**NN**	**0.65**	**3.34**	**0.00**	**0.53**	**0.52**
	DCPF	1.35	5.74	0.00	1.22	0.92

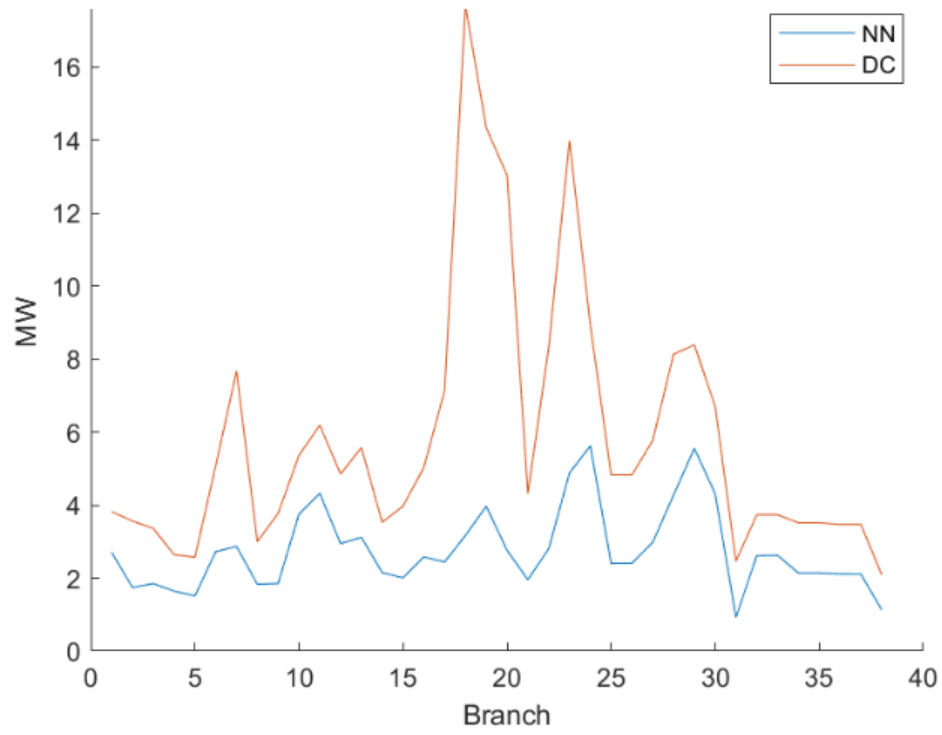


Fig. 8. Mean absolute value difference for active power flow of a 24-bus system.

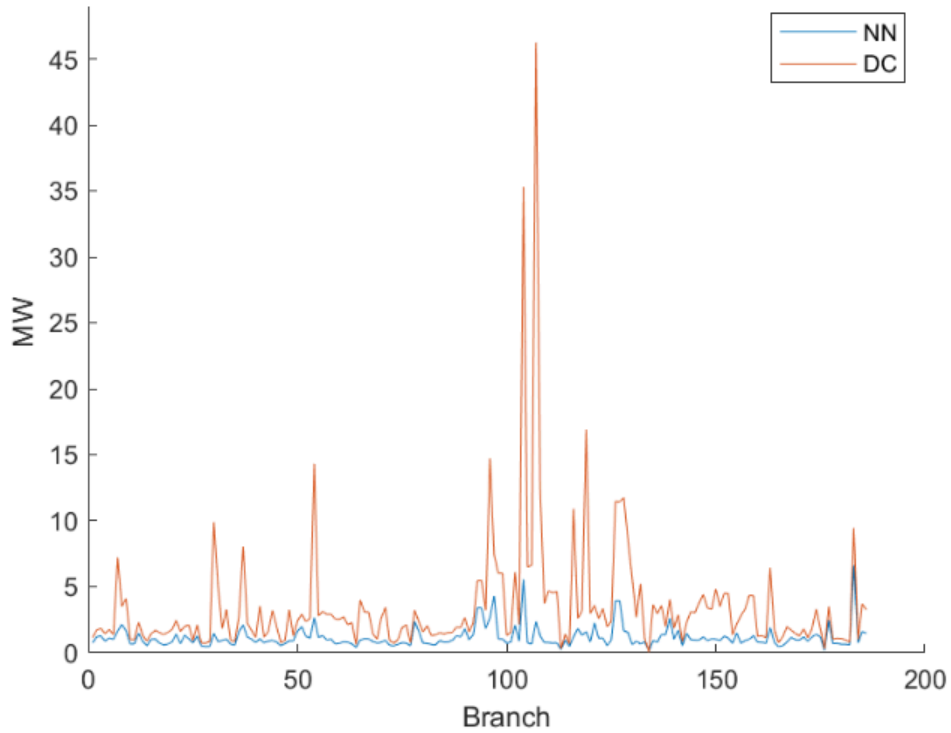


Fig. 9. Mean absolute value difference for active power flow of a 118-bus system.

Fig. 8 and Fig. 9 present the mean absolute error of line power flows on the IEEE 24-bus system and IEEE 118-bus

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验证误差率曲线与训练误差率曲线在图 7 中几乎完全吻合。训练好的神经网络模型似乎没有出现欠拟合或过拟合的问题。

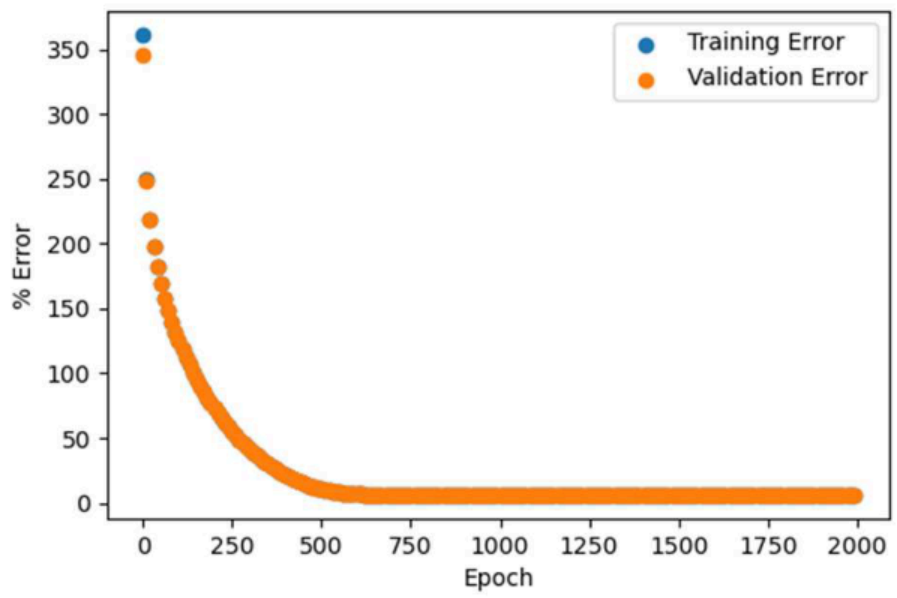


图 7. 24 节点系统有功功率潮流训练期间的平均绝对误差

所提出的神经网络模型能够高精度地预测多个系统中各母线的电压幅值：如表一所示，误差约为 0.5%。然而，由于几乎所有电压幅值都落在 0.9 标幺值到 1.1 标幺值的狭窄范围内，因此神经网络模型更容易学习和预测输出值。

表 I 所提出的 NN 模型在不同测试系统上的电压幅值平均 PRD。

电压幅值 系统百分比误差
9 将是 0.54
24 将是 0.60
39 将为 0.50
57 将为 0.33
118 路公交车 0.59

表 II 显示了所提出的神经网络模型和传统直流功率因数预测模型估算的线路潮流的绝对误差（单位：兆瓦）。所提出的神经网络模型预测的有功线路潮流比传统直流功率因数预测模型准确得多。在所有测量的统计值范围内，所提出的神经网络模型预测的线路潮流（兆瓦）都更接近完全精确的交流功率因数预测模型的参考值。

表 II 不同测试系统上所提出的 NN 模型与传统 DCPF 模型有功功率潮流的绝对值 (MW) 差异。

系统模型	平均值	最大值	最小值	中位数	标准差
9 将	NN	1.70	6.54	0.00	1.50	1.22
	DCPF	2.96	14.52	0.00	2.53	2.21
24 将	NN	2.77	21.60	0.00	2.26	2.27
	DCPF	5.92	43.38	0.00	4.29	5.53
39 将	NN	7.07	45.69	0.00	5.27	6.65
	DCPF	13.33	100.86	0.00	9.93	12.20
57 将	NN	0.54	11.08	0.00	0.27	0.73
	DCPF	1.65	23.19	0.00	0.96	1.98
118 将	NN	1.16	20.86	0.00	0.84	1.27
	DCPF	3.43	70.62	0.00	2.02	5.27

表 III 列出了仅与高度相关的结果。使用不同阈值水平对负载支路进行测试。DCPF 的结果不如所提出的神经网络模型准确。例如，对于 IEEE 24 节点系统，对于所有实际有功功率流超过 200 MW 的支路，所提出的神经网络模型的 PRD 均值仅为 0.65%，与完整 ACPF 模型报告的实际结果非常接近。然而，DCPF 模型的 PRD 为 1.35%，是所提出的神经网络模型误差的两倍。

表 III IEEE 24 节点系统上不同阈值水平下，所提出的 NN 模型和 DCPF 模型的有功功率潮流 PRD 比较。

系统模型	平均值	最大值	最小值	中位数	标准差
50 兆瓦	NN	1.72	17.14	0.00	1.26	1.62
	DCPF	6.42	75.28	0.00	3.43	10.02
100 兆瓦	NN	1.42	8.95	0.00	0.89	1.52
	DCPF	6.55	75.28	0.00	2.75	12.76
150 兆瓦	NN	1.33	8.95	0.00	0.76	1.59
	DCPF	2.96	24.63	0.00	1.63	3.53
200 兆瓦	NN	0.65	3.34	0.00	0.53	0.52
	DCPF	1.35	5.74	0.00	1.22	0.92

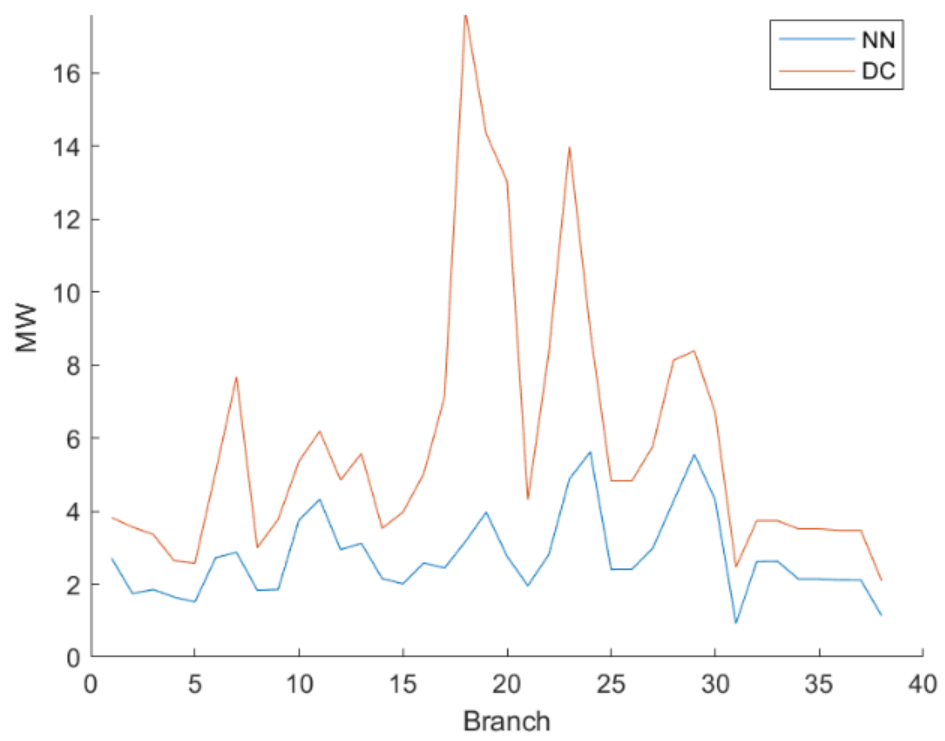


图 8. 24 节点系统有功功率潮流的平均绝对值差。

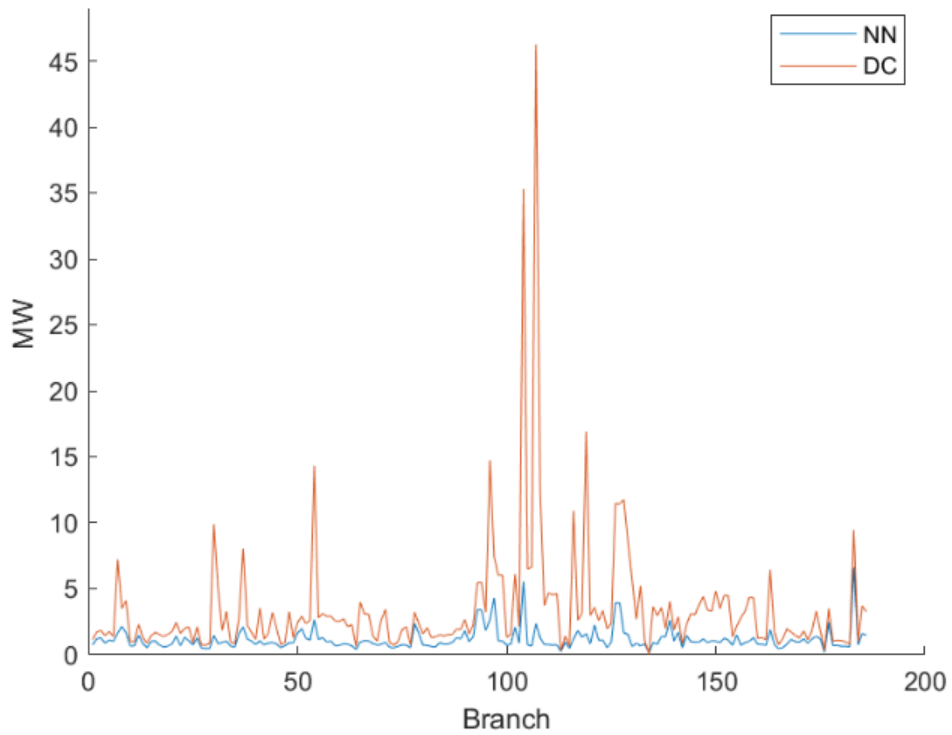


图 9. 118 节点系统有功功率潮流的平均绝对值差。

图 8 和图 9 分别展示了 IEEE 24 节点系统和 IEEE 118 节点系统中线路功率潮流的平均绝对误差。

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system respectively. From these two figures, it is observed that the proposed NN model outperforms the traditional DCPF model by a large margin. Note that the DCPF model has large outliers with great errors while the proposed NN model does not. For the proposed NN model, the predicted line active power flows are rarely more than 5MW away from the true values. It can be concluded that the proposed NN model is superior to the traditional DCPF model in terms of accuracy.

V. Conclusion

Based on the result, it can be concluded that the proposed NN model produces much better results at predicting line active power flows compared to the traditional DCPF model. In addition, the voltage magnitudes predicted by the proposed NN model closely match the results of ACPF. The potential of using the NN model as a supervised machine learning algorithm to calculate power flow has been explored in this paper. Unlike DCPF, no assumption must be made regarding voltage magnitudes and the proposed NN model can predict the values of voltage magnitudes very accurately. The values of line active power flows using the proposed NN model are very close to solutions solved by ACPF. In addition, solutions to both line active power flows and voltage magnitudes can be found quickly once the model has been trained offline. The proposed NN model performs faster with better results compared to the DCPF model.

Model selection is an important step for the ML algorithm. In addition, training the NN model may require a large data set of samples and computing resources. Building and training NN models can be difficult and computationally expensive, but the reward is significant. Applications of NN in power systems are still being explored and researched [1], [5]. Regardless, the potentials of using ML to build and improve upon previous power systems technologies are limitless.

VI. Future Work

The proposed NN model shows potentially useful applications of machine learning in power systems. Due to the adaptability nature of ML, NN predictions can be used as part of the optimization model to solve optimal power flow (OPF) for system with multiple renewable sources. With the NN model, we can directly calculate line flows as part of the constraint and simplify OPF calculation. Alternatively, we can also develop a different machine learning algorithm to account for unpredictability of renewable energy in OPF calculation. Another potential application of the proposed NN model is to quickly screen a long list of contingencies and filter out non-critical contingencies for large-scale power systems in real-time.

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系统分别如图所示。从这两幅图中可以看出，所提出的神经网络模型性能显著优于传统的直流潮流预测模型。值得注意的是，直流潮流预测模型存在较大的异常值，误差也较大，而所提出的神经网络模型则不存在这种情况。对于所提出的神经网络模型，预测的线路有功功率潮流与真实值之间的偏差很少超过 5MW。由此可以得出结论，所提出的神经网络模型在精度方面优于传统的直流潮流预测模型。

五、结论

根据结果可知，与传统的直流潮流计算（DCPF）模型相比，所提出的神经网络（NN）模型在预测线路有功功率方面表现更佳。此外，该 NN 模型预测的电压幅值与交流潮流计算（ACPF）的结果高度吻合。本文探讨了将 NN 模型作为监督式机器学习算法用于潮流计算的潜力。与 DCPF 不同，所提出的 NN 模型无需对电压幅值做任何假设，即可非常精确地预测电压幅值。使用所提出的 NN 模型计算得到的线路有功功率值与 ACPF 的计算结果非常接近。此外，一旦模型完成离线训练，即可快速求解线路有功功率和电压幅值。与 DCPF 模型相比，所提出的 NN 模型运行速度更快，结果更优。

模型选择是机器学习算法的重要步骤。
此外，训练神经网络模型可能需要大量的样本数据集和计算资源。构建和训练神经网络模型可能既困难又耗费计算资源，但回报也十分丰厚。神经网络在电力系统中的应用仍在探索和研究之中[1][5]。无论如何，利用机器学习来构建和改进现有电力系统技术的潜力是无限的。

六、未来工作

所提出的神经网络模型展示了机器学习在电力系统中的潜在应用价值。由于机器学习具有适应性，神经网络的预测结果可以作为优化模型的一部分，用于求解包含多个可再生能源系统的最优潮流（OPF）。利用该神经网络模型，我们可以直接计算线路潮流作为约束条件，从而简化 OPF 的计算。此外，我们还可以开发不同的机器学习算法来应对 OPF 计算中可再生能源的不可预测性。该神经网络模型的另一个潜在应用是快速筛选大量事故，并实时过滤掉大型电力系统中的非关键事故。

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